

Week 10 Discussion: Wiener Processes, Itô's Lemma, & Black–Scholes–Merton

Main focus (25min): (i) generalized Wiener processes + correlation; (ii) Itô processes + GBM/lognormality; (iii) universal application of BS equation to GBM and pricing formulas.

Module A) Generalized Wiener Processes & Correlation

Generalized Wiener (Brownian motion with drift):

$$X_t = X_0 + \mu t + \sigma B_t \quad \Longleftrightarrow \quad dX_t = \mu dt + \sigma dB_t,$$

so increments satisfy $X_{t+\Delta t} - X_t \sim N(\mu\Delta t, \sigma^2\Delta t)$.

Two processes and correlation: If

$$dX_1 = \mu_1 dt + \sigma_1 dB_1, \quad dX_2 = \mu_2 dt + \sigma_2 dB_2,$$

and $\text{Corr}(dB_1, dB_2) = \rho$, then

$$d(X_1 + X_2) = (\mu_1 + \mu_2) dt + (\sigma_1 dB_1 + \sigma_2 dB_2),$$

with **variance rate**

$$\sigma_{sum}^2 = \sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2 \quad (\text{so } \rho = 0 \Rightarrow \sigma_{sum}^2 = \sigma_1^2 + \sigma_2^2).$$

Practice A (Textbook 14.3): Sum of generalized Wiener processes

Variables X_1 and X_2 follow generalized Wiener processes with drift rates μ_1, μ_2 and variances σ_1^2, σ_2^2 . What process does $X_1 + X_2$ follow if:

1. the changes in X_1 and X_2 in any short interval are uncorrelated?
2. there is correlation ρ between the changes in any short interval?

TA Key for Practice A

Write $dX_i = \mu_i dt + \sigma_i dB_i$. Then $d(X_1 + X_2) = (\mu_1 + \mu_2)dt + \sigma_1 dB_1 + \sigma_2 dB_2$. So $X_1 + X_2$ is again a generalized Wiener process with

$$\text{drift rate } \mu_{sum} = \mu_1 + \mu_2.$$

Variance rate:

$$\text{Var}(dX_1 + dX_2) = \text{Var}(dX_1) + \text{Var}(dX_2) + 2\text{Cov}(dX_1, dX_2) = (\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2) dt.$$

Hence variance rate is $\sigma_1^2 + \sigma_2^2$ if $\rho = 0$, and $\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2$ in general.

Module B) Itô's Lemma & Geometric Brownian Motion (GBM)

Itô process: $dX_t = K_t dt + H_t dB_t$.

Itô's lemma (template): for $Z_t = g(t, X_t)$ with $g \in C^2$,

$$dZ_t = \left(g_t + g_x K_t + \frac{1}{2} g_{xx} H_t^2 \right) dt + g_x H_t dB_t.$$

GBM stock model: $dS_t = \mu S_t dt + \sigma S_t dB_t$.

Key identity (log trick):

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dB_t \quad \Rightarrow \quad \ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right).$$

So S_T is **lognormal**.

Practice B (Textbook 14.6): Does a portfolio follow GBM?

Stock A and stock B both follow geometric Brownian motion. Changes in any short interval of time are uncorrelated with each other. Does the value of a portfolio consisting of *one share of A and one share of B* follow geometric Brownian motion? Explain briefly.

TA Key for Practice B

Let S_t^A, S_t^B be GBMs driven by independent Brownian motions. Then each is lognormal. The portfolio value is

$$V_t = S_t^A + S_t^B.$$

A GBM has *lognormal* marginals: $\ln V_t$ must be normal. But V_t is a *sum of two lognormals*, which is generally **not** lognormal (no closed form lognormal law, and $\ln(S_t^A + S_t^B)$ is not normal). Therefore V_t does **not** follow GBM (except in degenerate special cases, e.g. one component is identically 0).

Module C) Risk-Neutrality, Universal BS Equation & Pricing Formulas

Risk-neutral modeling (no dividends): replace μ by $r \Rightarrow dS_t = rS_t dt + \sigma S_t dB_t^Q$.

Black-Scholes Formulas (European call/put, no dividends): with $\tau = T - t$,

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$

$$C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1).$$

Mini warm-up (Textbook 15.2): If annual volatility is $\sigma = 30\%$, then over 1 trading day ($\Delta t \approx 1/252$ year), $SD(\% \Delta S) \approx \sigma\sqrt{\Delta t} = 0.30/\sqrt{252} \approx 1.9\%$.

Applicability of the BS Equation: The Black-Scholes partial differential equation (PDE) applies to **ALL** derivative securities whose underlying asset follows a Geometric Brownian Motion (GBM).

Practice C (Textbook 15.16): BSM and Put-Call Parity

Show that the Black-Scholes-Merton formulas for call and put options satisfy put-call parity.

TA Key for Practice C

The Black-Scholes-Merton formulas at time $t = 0$ are:

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$$

$$P = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$$

We want to evaluate $C - P$:

$$C - P = [S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)] - [Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)]$$

Grouping the S_0 terms and Ke^{-rT} terms together:

$$C - P = S_0[\Phi(d_1) + \Phi(-d_1)] - Ke^{-rT}[\Phi(d_2) + \Phi(-d_2)]$$

A fundamental property of the standard normal cumulative distribution function $\Phi(x)$ (arising from the symmetry of the bell curve) is that the area to the left of x plus the area to the left of $-x$ exactly equals 1. That is:

$$\Phi(x) + \Phi(-x) = 1 \quad \text{for all } x.$$

Applying this property to d_1 and d_2 :

$$C - P = S_0(1) - Ke^{-rT}(1) = S_0 - Ke^{-rT}$$

This exactly matches the standard put-call parity relationship for non-dividend-paying stocks.